

Supersymmetric links, stable source vacua and the CKM boundary

An isolated alignment branch, physical weak CP and an exact
orthogonal-frame obstruction

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Joint local vacua are calculated without a nominated angle. The orthogonal axis branch fails the
observed hierarchy. Global selection, operator protection and a golden CKM relation remain
open.

This calculation connects the previously separate link, scalar and quark constructions in a specified effective field theory. It derives an isolated group-aligned supersymmetric link branch and its complete local mass spectrum. It then proves that the minimal exact-supersymmetry source sector has only zero flavon vacua. A specified spurion-breaking extension supplies locally stable nonzero sources and a joint vacuum with physical weak CP violation. Finally an exact finite census excludes every orthogonal axis branch from realistic CKM mixing, and a permitted cross-sector interaction moves the solved quark frame.

Neither the golden coefficient nor 66 degrees enters the superpotential, source directions, breaking terms or vacuum parameters. The positive result is an explicit dynamical realization of discrete alignment and weak CP on a locally stable branch. The negative result is a sharp limitation of that branch, not a no-go theorem for broken Valentiner flavor models. The global family spectrum from M22 is not used as a physical input or claimed to be unified here.

One specified flavor effective theory

Take $G_u \times G_d$, with each factor the exact $3.A_6$ triplet group. The chiral link K transforms as $(\bar{3}_u, 3_d)$. There are three source flavons Φ_{ui} in $(\bar{3}_u, 1)$ and three Φ_{di} in $(1, \bar{3}_d)$. Each source has its own C_6 shaping charge; its corresponding right-handed conjugate quark has the opposite charge. All other family messengers in the displayed quark chain are neutral under these factors. The left-handed quark doublet Q transforms as $(3_u, 1)$. Two Higgs doublets and vectorlike quarks carry the usual Standard Model gauge representations. Matter parity and baryon conservation are assumed. The canonical kinetic metric, finite-order EFT truncation and eventual gauge/anomaly completion remain separate assumptions.

Write $F_{6\bar{}}$ for the complex-conjugate polynomial coefficients of the previously derived Valentiner sextic, and $I_{6\bar{}}$ for the corresponding holomorphic link invariant. On the link-and-source scalar sector, all nonconstant holomorphic invariants through field degree six give

$$W_{\text{flavor}} = a \det(K) + b I_{6\bar{}}(K) + c \det(K)^2 + \sum_i \kappa_i F_{6\bar{}}(\Phi_i).$$

There are nine independent contractions: the link determinant, its square, the new link sextic and six source sextics. Individual shaping charges exclude mixed link/source and mixed-source holomorphic terms at these degrees. This is a complete statement about this scalar superpotential sector, not a complete Kähler, breaking or higher-degree quark EFT enumeration. CP permits real coefficients in the checked convention. The source coefficients are independently nonzero; no golden relation among them is assumed. With fields of mass dimension one, a is dimensionless and b , c and κ have dimension minus three.

The physical Dirac row link is $L = \text{conjugate}(K)$. The quark superpotential contains $Q.H_u U^c$, $\mu_u U.U^c$, source terms $u_i^c \Phi_{ui}^T U$, $Q.H_d D_u^c$, $\mu_d D_u.D_u^c$, $m_d D_d.D_d^c$, the forward term $g D_u^T K D_d^c$, and source terms $d_i^c \Phi_{di}^T D_d$. All column couplings are independent real inputs. A Higgs μ term and positive Higgs breaking masses keep the Higgs backgrounds zero in this flavor-scale calculation; electroweak breaking is not derived.

The earliest holomorphic reverse vertex is $\eta D_d^T \text{adj}(K) D_u^c$. It has field degree four, with an independent coefficient of mass dimension minus one. The adjugate is a quadratic polynomial in K , and is defined even at singular K . It replaces the invalid attempt to put K -dagger in a superpotential with one chiral link. The corresponding physical heavy down-quark block is $M = [[\mu, g, L], [\eta \text{adj}(L), m_d, I]]$. Canonical tree matching is applied to this block, rather than copying the earlier nonsupersymmetric result. Additional higher-degree quark terms are not asserted absent.

The group-aligned branch and all nine link masses

$$r^3 = -\frac{a}{32b+2c}, \quad m_1 = |96b + 6c| |r|^4, \quad m_8 = |72b| |r|^4$$

The exact symmetric tensor T has norm squared 16. Its two-index reduced tensor is $(4/3)(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$; all 81 entries are checked in the coefficient field. This gives $d^2 I_6(I)[E,F] = 40(\text{Tr}(E)\text{Tr}(F) + \text{Tr}(EF))$. Combined with the determinant derivatives, it establishes the F-flat branch

$r^3 = -a/(32b+2c)$, with $K = r \text{conjugate}(R(g))$.

At a nonzero r , every group transform has zero F energy. On that branch the superpotential Hessian is $r^4[(8b+2c)\text{Tr}(E)\text{Tr}(F) + 72b \text{Tr}(EF)]$. The nine complex link modes split into a singlet with fermion mass $|96b+6c||r|^4$ and eight modes with mass $|72b||r|^4$. Canonical scalar squared masses are their squares, with real multiplicities two and sixteen. Thus a nonzero branch is locally isolated in the link sector when b and $16b+c$ are nonzero, in addition to the existence conditions. No continuous family group is gauged here, so these eight massive directions are not Goldstone modes.

For $a = -.0324$, $b = 1$ and $c = .2$, the real branch has $r = .1$. Its fermion masses are .00972 and .0072. Four independent coefficient variations retain the nonzero branch and positive masses. The radius follows the coupling ratio; it is not a mixing-angle input. The origin also remains F-flat. Other F-flat link solutions and global occupation have not been classified.

There is a local protection statement worth separating from a CKM prediction. The diagonal family stabilizer of rI fixes only the scalar matrices, because $3 \text{ tensor } 3\text{bar}$ decomposes as $1+8$. An invariant superpotential gradient at rI must therefore lie in that one-dimensional fixed subspace. With a nonzero singlet Hessian, small invariant holomorphic deformations preserve a nearby scalar-radius branch by the implicit function theorem. They do not need the specially chosen coefficients of the original angular potential. This protects a local link alignment branch; source-dependent operators can remove that stabilizer and change the relative quark frame.

What exact supersymmetry does to the sources

$$\nabla F_6(\phi) = 0 \iff \phi = 0$$

The minimal source superpotential does not stabilize nonzero sources. The exact computation establishes $\text{grad } F_6(\phi) = 0$ if and only if $\phi = 0$. To verify this, write $u = x^2$, $v = y^2$ and $w = z^2$, turning F_6 into a cyclic cubic H . One-coordinate support has a nonzero derivative. The two-coordinate derivative resultant is $(67797 - 35235 i \sqrt{15})/32$, which is nonzero. On three-coordinate support, the two-variable Gröbner basis of H_u , H_v and H_w after $u=1$ is [1]. Cyclicity covers every projective patch. The independent test works directly with the original quintic derivatives, without the squared-coordinate reduction, and also obtains [1]. Conjugating the representation leaves the obstruction unchanged.

Consequently all six flavons vanish in the exact-SUSY, zero-matter, zero-Higgs vacuum slice for nonzero κ_i . Both Yukawa matrices then vanish. There is no CKM frame to extract. The eighteen complex source modes have zero quadratic mass, but their potential is lifted at degree ten; they are not flat moduli. This is an obstruction to this minimal field-content/truncation, not to models with driving fields, different charges or breaking interactions.

A specified breaking extension with stable nonzero sources

$$V_\phi = \kappa^2 \|\nabla F_6\|^2 - m^2 \|\phi\|^2 + 2A\kappa \operatorname{Re} F_6$$

Add for each source $V_i = \kappa_i^2 \operatorname{norm}(\operatorname{grad} F_6)^2 - m_i^2 \operatorname{norm}(\Phi_i)^2 + 2A_i \kappa_i \operatorname{Re} F_6(\Phi_i)$. These are explicitly specified symmetry-respecting spurion-breaking EFT terms. They are not claimed to exhaust the allowed breaking action, and their origin or common coefficients are not derived. The independent column labels and their quark couplings are already field-content inputs, not a derivation of three families.

There is a stationary axis branch $\Phi = \rho e_j$ when $m_i^2 = \kappa_i^2 \rho^8 (180 + 6t)$, where $t = A_i / (\kappa_i \rho^4)$. The two transverse tensor coefficient magnitudes are exactly $\sqrt{45/2}$. In canonical real coordinates the six squared masses, divided by $\kappa_i^2 \rho^8$, are $1440 + 24t$, $-36t$, and each of $-90 - 6t$ plus/minus $(3\sqrt{10}/2)|48 + 2t|$ repeated twice. They are positive with positive m_i^2 throughout

$-30 < t < -(90 + 72\sqrt{10})/(6 + 3\sqrt{10})$.

At $t = -24$ they are 864, 864 and four copies of 54. The source vacuum energy is $-48 \kappa_i^2 \rho^{10}$, below the source origin. Every group transform of an axis has the same energy and spectrum. The $t = -26$, -24 and -22 cases are retained; direct six-dimensional real Hessians independently confirm their positive spectra. These are locally stable vacua. Other source extrema and the global winners are not classified.

Nonzero source F backgrounds also induce bilinear terms in the quark scalar potential. They cannot be ignored in a stability claim. At $\rho = .1$, $\kappa = 1$ and the stated real quark column couplings, their operator norm is at most $5.4e-5$. Adding a specified positive universal quark scalar breaking mass squared of $1e-4$ gives a rigorous lower bound $4.6e-5$ after including the nonnegative supersymmetric mass block. Explicit up and down scalar matrices have smallest squared masses approximately $9.99965e-5$ and $9.99985e-5$. Positive Higgs masses complete local stability in the declared zero-Higgs, zero-matter slice. This is not a threshold- or loop-matched complete Standard Model vacuum.

CP on a link and on a joint vacuum

For real CP-compatible coefficients, r^3 is real. The three cube-root phases differ by a central element of $3A_6$. The group link states form one family orbit. Acting with the elementary CP involution sends $rR(g)$ to $r^*R(u(g))$; a family transformation and, if needed, a central element restore the vacuum. The group link by itself therefore conserves a CP operation combined with family symmetry. Complex entries alone do not establish weak CP.

The nonzero source vacua change that conclusion. Exact field arithmetic identifies 45 axis rays, 30 unordered orthogonal frames, and two family orbits of fifteen frames each. The generator action on exact rank-one projectors, their orthogonality and the orbit partition are checked without clustering tolerances. Their 900 ordered up/down frame pairs cover every group link choice, because a group link permutes the ray and frame sets. Mass-order permutations only permute the displayed magnitudes or reverse the sign of J .

A joint stationary vacuum from the retained frame pairs has $J = -\sqrt{3}/32$, with an exactly computed invariant-product imaginary part. Canonically normalized quark matrices reproduce $J = -.0541265877$ and their CP conjugates give the opposite sign, with identical spectra and magnitudes. An independent weak-basis commutator determinant is nonzero. The worked pair has $|V_{us}| = |V_{cb}| = 1/2$, $|V_{ub}| = 1/\sqrt{2}$, and $C_{\text{eff}} = \sqrt{2}$. The frame pair was selected as the

first nonzero-J witness, not by an energy preference. Its up and down spectra come from independently chosen real column couplings. This demonstrates physical weak CP in a locally stable branch; it does not select one sign or reproduce observed quark mixing. Strong CP is not solved.

A complete obstruction for the orthogonal axis branch

$$|V_{ij}| = 0 \quad \text{or} \quad |V_{ij}| \geq \frac{\sqrt{5}-1}{4} \approx 0.309$$

Every source column in this branch is a group-rotated coordinate ray. An overlap of two such rays is an entry of a group matrix, since its relative group product remains in 3.A6. The implementation enumerates all exact squared entry values and compares them using a rational enclosure of $\sqrt{5}$. The smallest nonzero squared value is $(3-\sqrt{5})/8$, hence the smallest nonzero mixing magnitude is $(\sqrt{5}-1)/4 = .3090169944$. The ordering is certified by rational arithmetic, not inferred from a numerical minimum.

The entire orthogonal axis branch therefore has entries either zero or at least .3090169944. It cannot yield the observed nonzero $|V_{ub}|$ near .004, under any up/down frame pair, group link or mass permutation. This is considerably stronger than rejecting the displayed CP witness. It is a dynamical realization of a residual-frame obstruction: the scalar model has now produced the relevant stationary directions and positive local masses. Nonorthogonal source choices and other source extrema lie outside the obstruction and require their own joint-vacuum calculations.

The permitted interaction still moves the solved frame

$$H \Delta q = -\nabla O, \quad O = |\phi_{u1}^\dagger L \phi_{d1}|^2$$

The scalar EFT permits epsilon $|\phi_{u1}^\dagger L \phi_{d1}|^2$ under both family factors, all six shaping charges and CP. At the solved CP witness, the first source overlap probability is 1/4. Its constrained source torque is nonzero, with squared norm $r^4 \rho^6 B(1-B) = 1.875e-11$ in the declared units. Thus this interaction acts on an actual stable vacuum rather than on an assumed source frame.

The full 54-real-dimensional link/source Hessian is positive, with smallest eigenvalue about $5.4e-7$. Solving $H \delta x = -\text{grad } O$ gives a first-order stationary-vacuum response. Canonical quark matching along that response changes both magnitudes and weak CP: $dC_{\text{eff}}/d\epsilon$ is approximately 9.616 and $dJ/d\epsilon$ approximately .03590. These numbers use a numerical Hessian and dimensionful EFT normalization; they are not experimental error bars, a radiative coefficient estimate or a naturalness scale. The retained epsilon plus/minus $1e-6$ cases are linear continuations, not freshly minimized finite-epsilon vacua.

A mechanism protecting a precise relative frame must consequently constrain this operator or account for its effects. The product family group by itself does not do so. Generic small deformations preserve a nearby local minimum, but they need not preserve its CKM quantities exactly. This distinguishes stability of a vacuum from protection of an observable relation.

What the push establishes and what should follow

The earlier scalar-matching gap is now closed for a specified EFT: the determinant and sextic give a locally isolated nonzero link branch; a specified breaking sector gives nonzero sources and positive local spectra; a holomorphic cofactor replaces the conjugate-link reverse vertex; and physical weak CP is calculated from canonical quark matrices at a joint vacuum. The branch still fails the hierarchy and permits an explicit observable-changing cross-sector interaction. No target angle was supplied to obtain these conclusions.

The next useful search should explore nonorthogonal source vacua or new joint source/link contractions, while retaining the complete allowed breaking and kinetic operator spaces. It must derive their coefficients or demonstrate robustness across the independent ones. Adding a driving field or another representation is justified only if its operator census actually removes a demonstrated obstruction. The present calculation supplies concrete tests for that decision and avoids confusing another constructed golden root with a flavor prediction.

Reproduction and relation to established work

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_valentiner_susy_vacua.py`. Its scientific receipt records exact tensor identities, the source critical-locus reduction, parameter variations, the finite frame census, canonical CP pairs, scalar stability bounds and the allowed-response calculation. Run `python python/tests/test_valentiner_susy_vacua.py` for independent original-polynomial, Hessian, cofactor covariance, holomorphy, full fermion-mass and CP-orbit checks. Numerical and exact statements are identified separately. No new Lean theorem or experimental fit is claimed.

The classical Valentiner sextic and the general idea of supersymmetric flavon alignment are established mathematics and model-building tools. This draft does not claim their discovery. Primary background: S. P. Martin, A Supersymmetry Primer, <https://arxiv.org/abs/hep-ph/9709356> ; F. Feruglio, C. Hagedorn and L. Merlo, Vacuum Alignment in SUSY A4 Models, <https://arxiv.org/abs/0910.4058> ; C. Hagedorn, M. L. Lopez-Ibanez, M. J. Perez, M. H. Rahat and O. Vives, Flavon vacuum alignment beyond SUSY, <https://arxiv.org/abs/2312.07430> ; M.-C. Chen et al., CP Violation from Finite Groups, <https://arxiv.org/abs/1402.0507> . The cited alignment papers concern their own models, not a proof of our coefficients or CKM relation. The exact branch, spectra, finite census and response reported here are calculations for this declared construction. Publication novelty requires comparison and independent review.